Psychon Bull Rev (2015) 22:1054–1061
DOI 10.3758/s13423-014-0752-9

BRIEF REPORT

Language proficiency and morpho-orthographic segmentation

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Published online: 28 October 2014

Abstract One key finding in support of the hypothesis that written words are automatically parsed into component morphemes independently of the true morphological structure of the stimuli, so-called morpho-orthographic segmentation, is that suffixed nonword primes facilitate the visual recognition of a stem target (*rapidifier-RAPIDE*) whereas non-suffixed primes (*rapiduit-RAPIDE*) do not. However, Morris, Porter, Grainger, and Holcomb (Language & Cognitive Processes, 26(4–6), 558–599, 2011)reported equivalent priming from suffixed and non-suffixed nonword primes, hence questioning the morphological nature of prior findings. Here we provide a further investigation of masked priming with morphologically complex nonword primes with an aim to isolate factors that modulate the size of these priming effects. We conducted a masked primed lexical decision experiment in French, in which the same target (*TRISTE*) was preceded by a suffixed word (*tristesse*), a suffixed nonword (*tristerie*), a non-suffixed nonword (*tristald*), or an unrelated prime word (*direction*). Participants were split into two groups, based on their language proficiency. The results show that in the high proficiency group, comparable magnitudes of priming were obtained in all three related prime conditions (including the non-suffixed condition) relative to unrelated primes, whereas in the low proficiency group, priming was significantly reduced in the non-suffixed condition compared to the two suffixed conditions. These findings provide further evidence that individual differences in language proficiency can modulate the impact of morphological factors during reading, and an explanation for the discrepant findings in prior research.

Keywords Morphological processing · Morpho-orthographic segmentation · Language proficiency · Lexical decision · Masked priming

Much recent research has been dedicated towards understanding how morphologically complex words are processed during visual word recognition. One key finding, obtained using masked priming combined with the lexical decision task, is that the recognition of a target word is facilitated when preceded by a related morphologically complex word (*painter-PAINT*) or a word with a pseudo-morphological structure (*corner-CORN*), relative to non-morphological controls (*cashew-CASH*). Such evidence suggests that not only suffixed (*painter*) but also pseudo-suffixed words (*corner*) are rapidly decomposed into their *morpho-orthographic* subunits (*paint+er; corn+er*) at early stages of visual word recognition (for a review, see Rastle & Davis, 2008), and points to an initial morpho-orthographic segmentation process that is sensitive to surface morphological structure independently of the true morphological status of the constituents (Diependaele, Sandra, & Grainger, 2009; Longtin, Segui, & Hallé, 2003; Rastle, Davis, & New, 2004; Taft, 2004).

Particularly convincing evidence for rapid, automatic morpho-orthographic segmentation comes from a number of studies using morphologically structured *nonwords* as primes. Longtin and Meunier (2005) carried out a masked priming study in French in which primes were always nonwords. They compared semantically interpretable (*rapidifier–RAPIDE*) and non-interpretable (*sportation–SPORT*) nonword primes, and found similar-sized effects relative to a non-morphological control (*rapiduit-RAPIDE*, where "uit" is

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not a suffix), suggesting that morphological segmentation occurs for all morphologically structured items, even if they are not words (for related evidence from Spanish, see also Beyersmann, Duñabeitia, Carreiras, Coltheart, & Castles, 2013). McCormick, Rastle, and Davis (2009) reported similar effects in English, demonstrating that morphologically complex nonwords with orthographic alterations in the stem (e.g., *adorage–ADORE*) produced significant priming of the stem-target. Such priming effects constitute particularly strong evidence for fast-acting, automatic morpho-orthographic segmentation processes operating on any string of letters composed of a stem and an affix, independently of the lexical status of the string and the legality of the morpheme combination. They point to the existence of sublexical orthographic representations of stems and affixes, and a mechanism that enables isolation of the stem representation whenever it is accompanied by an affix (Taft & Forster, 1975).

As pointed out by McCormick et al. (2009), the key finding in these studies is that complex nonword primes facilitate recognition of embedded stem targets more so than matched orthographic control primes. Ideally, this should be demonstrated with the same targets (e.g., *adorage-ADORE* vs. *adorige-ADORE*) and the same participants. However, none of the experiments reported in Longtin and Meunier (2005) and McCormick et al. (2009) involved such a comparison. This is problematical given that Morris, Porter, Grainger, and Holcomb (2011) reported equivalent behavioral priming effects from suffixed (*flexify-FLEX*) and non-suffixed nonword primes (*flexint-FLEX*), leaving open the question as to the robustness of complex nonword priming.

Given the theoretical importance of priming effects obtained with complex nonword primes, we sought to replicate these findings using French stimuli (as in the Longtin & Meunier, 2005, study) and a within-item design (as in the Morris et al., 2011, study). To this end, the same target stimulus (*TRISTE*) was coupled with a suffixed word prime (*tristesse* [sadness]), a suffixed nonword prime (*tristerie* [sadation]), a non-suffixed nonword prime (*tristald* [sadald]), and an unrelated word prime (*direction* [direction]). Crucially, we also manipulated several other factors with an aim to isolate possible reasons for the discrepant findings in prior research. These factors were: affix frequency, language proficiency of the participants, and type of nonword stimuli used in the lexical decision task.

Language proficiency, in particular, would appear to be a good candidate for explaining the divergent findings, given recent evidence that individual differences in spelling ability and vocabulary determine the magnitude of morphological priming effects (Andrews & Lo, 2013). Good spellers with comparatively poor vocabulary showed comparable magnitudes of morphological (*painter-PAINT*) and pseudo-morphological (*corner-CORN*) priming, whereas poor spellers with comparatively good vocabulary showed significantly greater morphological relative to pseudo-morphological priming. Following Andrews and Lo (2013), we therefore measured the spelling ability and vocabulary of our participants. We also manipulated suffix frequency, given the possibility that complex nonword priming might be more robust when the nonword primes are composed of a stem and a high-frequency suffix. Finally, given prior evidence that increasing the difficulty of lexical decisions by increasing the similarity of the nonword targets to real words can increase the depth of processing of word targets (e.g., Grainger & Jacobs, 1996; Stone & Van Orden, 1993), we manipulated the orthographic similarity of the nonword targets to real words. The logic here is that complex nonword priming might depend on the depth of processing of target words, and therefore that subtle differences in the type of nonword targets used in prior research might be one source of the discrepant findings.

Method

Participants

Forty-eight students from Aix-Marseille University, all French native speakers, participated for monetary reimbursement (€5/30 minutes).

Materials

Fifty word targets were selected from Lexique (New, Pallier, Brysbaert, & Ferrand, 2004). Each target was preceded by a suffixed word prime (*tristesse-TRISTE* [sadness-SAD]), a suffixed nonword prime (*tristerie-TRISTE* [sadation-SAD]), a non-suffixed nonword prime (*tristald-TRISTE* [sadald-SAD]), and an unrelated prime (*direction-TRISTE* [direction-SAD]). Suffixed word primes were real words comprising stem (*triste*) and suffix (*esse*). Each suffix occurred in five different word-contexts (e.g., *esse* occurred in *tristesse, jeunesse, sagesse, vitesse,* and *hôtesse*). Suffixed nonword primes were created using the same stem, but a different suffix of comparable frequency (*rie*), such that the whole letter string was not a word. Non-suffixed nonword primes were created by combining the stem with a non-morphemic ending (*ald*). Each non-morphemic ending occurred in five different contexts. Unrelated primes were suffixed words and all letters different from the target. All nonwords were orthographically legal and pronounceable. The four prime conditions were matched on length (Appendix 1).

Trials were divided into two subsets: one where primes comprised high-frequency suffixes (*−ette*, *−ier*, *−eur*, *−ion*, and *-age*) and one where primes comprised low-frequency suffixes (−*esse*, −*oir*, −*ade*, −*ien*, and -*rie*). The mean word-final position frequency (WFPF) for low-frequency suffixes was 109.8 and was 153.8 for high-frequency suffixes. Suffixed words and nonwords used the same suffixes, but with different stems. Non-suffixed nonwords were created using five high-frequency (−*nie*, −*ire*, −*ide*, −*ert*, and -*use* [WFPF: 167.8]) and five low-frequency non-morphemic endings (−*uor*, −*ald*, −*abe*, −*uto*, and -*bli* [WFPF: 4.4]). High and low frequency conditions were matched on length, frequency, suffix length, and non-morphemic ending length.

Two different sets of 50 nonword targets were created in a between-participant manipulation of the orthographic similarity of nonwords to real words. Set 1 (presented to participant group A) was created by replacing three letters in a baseword. Primes preceding nonword targets were created based on the same principles as those preceding word targets. Nonword and word targets, as well as their corresponding primes, were matched on length. To avoid participants seeing any target twice, we created four counterbalanced lists. In Set 2 (presented to participant group B), the orthographic similarity of nonword targets to real words was increased by changing only one letter in the baseword (*froin* vs. *froid*). As a result, the average number of orthographic neighbours (Coltheart's N) was significantly higher in Set 2 than in Set 1 (4.77 [SD: 5.18] vs. 1.96 [SD: 4.56], p<.001).

Procedure

Stimuli were presented in the centre of a CRT (cathay ray tube) computer screen, using DMDX (Forster & Forster, 2003). Each trial consisted of a 500-ms forward mask of hash keys, then a 50-ms prime in lowercase, then the uppercase target. The target remained present until the response or until 3 seconds had elapsed. Participants were instructed to respond as quickly and accurately as possible.

Measures of individual differences

Upon completion of the masked primed lexical decision task, each participant was assessed with a spelling recognition, spelling dictation, and vocabulary proficiency test.

Spelling recognition Participants performed the French LEXTALE (Brysbaert, 2013), which served as a spelling recognition test. Participants were asked to identify correctly spelled words within a list of 84 target items, consisting of 56 words and 28 misspelled nonwords. The score was the number of correct word responses from which the doubled number of false alarms were subtracted.

Spelling dictation Following Andrews and Hersch (2010), we created a spelling dictation test in French, consisting of 20 words, which we selected from Lexique (New et al., 2004). All words were adjectives or nouns, eight -letters long, low-frequency (mean: 0.97) with few orthographic neighbors (mean: 0.4). Words were read out aloud by the experimenter. The score was the number of correctly spelled words. Existing homophones of the target (e.g., silent plural-s: *phobique/phobiques*) were counted as correct. The test's materials are listed in Appendix 2.

Vocabulary The WAIS-III Vocabulary subtest was administered to obtain an estimate of general level of lexical knowledge (Wechsler, 1997). In this test, participants were presented with individual words and asked to provide detailed definitions. The score was calculated based on the WAIS-III scoring system.

Results

Lexical decisions to word targets were analyzed as follows. Incorrect responses were removed from the reaction time analysis (2.6 % of all data). Two target words were removed, because error rates were above 30 %. Reaction times (RTs) were logarithmically transformed and outliers trimmed using interquartile trimming. Transformed RTs smaller than Q1–(2.5×IQR) or larger than Q3+(2.5×IQR), by either participants or items, were excluded from the analyses (Q1=first quartile, Q3=third quartile, and IQR=Q3–Q1=interquartile range). This trimming led to the removal of 2.8 % of the data. RTs and error rates were analyzed for each subject (see Table 1).

Table 1 Reaction times (in ms) and error rates (in %), averaged across items for each participant. Standard deviations are presented in parentheses

Condition	Reaction times	Error rates	Example
All participants			
Suffixed word	531 (85)	0.9 (3.4)	*tristesse-TRISTE*
Suffixed nonword	542 (99)	2.8 (6.9)	*tristerie-TRISTE*
Non-suffixed nonword	547 (92)	3.5 (5.9)	*tristald-TRISTE*
Unrelated	573 (103)	1.7 (5.0)	*direction-TRISTE*
High language proficiency			
Suffixed word	532 (81)	0.0 (0.0)	*tristesse-TRISTE*
Suffixed nonword	539 (78)	2.1 (7.4)	*tristerie-TRISTE*
Non-suffixed nonword	535 (74)	2.5 (4.1)	*tristald-TRISTE*
Unrelated	570 (84)	1.3 (2.9)	*direction-TRISTE*
Low language proficiency			
Suffixed word	530 (90)	1.9 (4.6)	*tristesse-TRISTE*
Suffixed nonword	545 (117)	3.4 (6.3)	*tristerie-TRISTE*
Non-suffixed nonword	559 (107)	4.6 (7.1)	*tristald-TRISTE*
Unrelated	575 (120)	2.2 (6.6)	*direction-TRISTE*

We used linear mixed-effect modelling to perform the main analyses (Baayen, 2008; Baayen, Davidson, & Bates, 2008). Fixed and random effects were only included if they significantly improved the model's fit in a backward stepwise model selection procedure. Models were selected using chi-squared log-likelihood ratio tests with regular maximum likelihood parameter estimation. Trial order was included to control for longitudinal task effects such as fatigue or habituation. To assess whether the obtained effects were modulated by individual differences of spelling ability and vocabulary, the spelling recognition, spelling dictation, and vocabulary scores were standardized. A composite measure of spelling (SpellZ) was calculated by averaging the standard scores of the two spelling tests. The composite SpellZ measures and standardized vocabulary scores (VocabZ) were highly correlated (r=.561). Following Andrews and Lo (2013), two orthogonal measures of individual differences were used. The first one (LanguageProficiencyZ) captured the common variance between SpellZ and VocabZ, and had high positive correlations with both measures (spelling: r=.938; vocabulary: r=.813). The second one (SpellVocabDiffZ) captured the difference between the two measures, and showed opposite directions of relationship with spelling (r=.322) and vocabulary (r=−.602).

A linear mixed-effects model as implemented in the lme4 package (Bates, Maechler, Bolker, & Walker, 2014) in the statistical software R (Version 3.0.3; RDevelopmentCoreTeam, 2008) was created with six fixed effects factors (primetype: suffixed-word, suffixed-nonword, non-suffixed-nonword, unrelated word; suffix-frequency: high, low; nonword neighborhood density: low, high; LanguageProficiencyZ; SpellVocabDiffZ; trial order), their interactions, and two random effects factors (random intercepts for subjects and items). P-values were determined using the package lmerTest (Kuznetsova, Brockhoff, & Christensen, 2014). RT analyses revealed that SpellVocabDiffZ did not interact with any priming effects. Critically, however, LanguageProficiencyZ interacted significantly with non-suffixed priming (t=2.14, p=.033), but not with suffixed word and suffixed nonword priming (t=1.10, p=.274; t=0.96, p=.336; see Fig. 1). There was also a significant effect of primetype, showing that participants responded faster to targets preceded by suffixed words, suffixed nonwords and non-suffixed nonwords relative to the unrelated condition (t=7.65, p<.001; t=5.88, p<.001; t=4.68, p<.001, respectively). Moreover, participants responded faster to targets preceded by suffixed words than non-suffixed nonwords (t=2.93, p=.004). The effect of trial order was significant (t=3.57, p<.001). No other effects were significant.

Participants were then split into two groups depending on whether their language proficiency score fell above (high proficiency group) or below (low proficiency group) the median LanguageProficiencyZ value. RT analyses of the high proficiency group revealed a significant effect of primetype, showing that participants responded faster to targets preceded by suffixed words, suffixed nonwords, and non-suffixed nonwords relative to the unrelated condition (t=5.34, p<.001; t=4.16, p=.001; t=4.61, p<.001). Interestingly, the results of the low proficiency group revealed a different pattern. As in the high proficiency group, suffixed word, suffixed nonword, and non-suffixed nonword conditions differed significantly from the unrelated condition (t=5.59, p<.001, t=4.34, p<.001 and t=2.12, p=.034). Critically, however, the magnitude of priming in the suffixed word and suffixed nonword conditions was significantly larger than in the non-suffixed condition (t=3.43, p<.001 and t=2.20, p=.028), while there was no difference between the two suffixed conditions (t=1.23, p=.218). There was also a significant effect of trial order (t=2.48, p=.013). No other effects were significant.

Error analyses followed the same logic as the RT analyses. We applied a binomial variance assumption to the trial-level binary data using the function *glmer* as part of the R-package *lme4*. Participants made more errors responding to targets in the suffixed and non-suffixed nonword conditions than in the suffixed word condition (t=2.41, p=.016; t=3.04, p=.002),

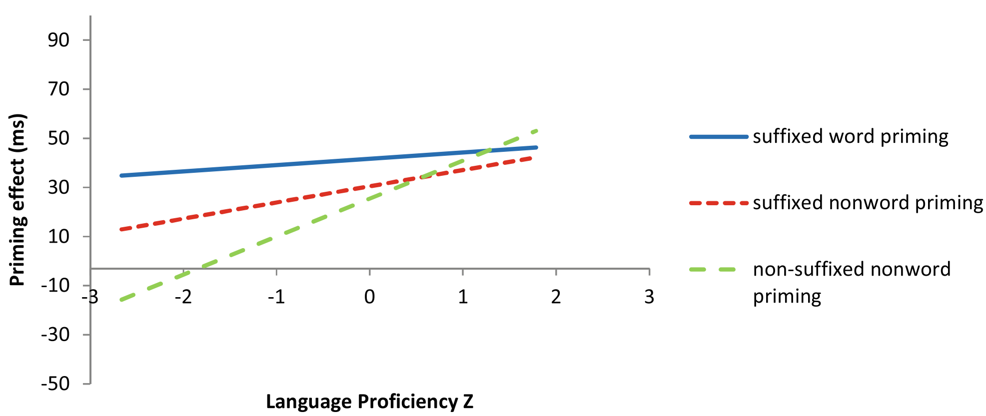

Fig. 1 Priming effects for targets preceded by suffixed word, suffixed nonword, and non-suffixed nonword primes (relative to the unrelated control condition), as a function of individual differences in language proficiency

and more errors responding to targets preceded by non-suffixed nonwords than unrelated words (t=2.09, p=.037). No other effects were significant.

Discussion

The main aim of the present study was to replicate a key finding in the masked morphological priming literature, first reported by Longtin and Meunier (2005), showing that suffixed nonword primes facilitate the recognition of embedded stem targets, whereas non-suffixed nonwords do not. Given a recent failure to replicate these findings (Morris et al., 2011), our goal was to use stimuli similar to Longtin and Meunier but in the context of an improved methodological design (using the same targets across all conditions). We further sought to examine whether differences in affix frequency, individual language proficiency, and nonword neighbourhood density might be the source of these divergent findings.

Across all participants, we failed to replicate Longtin and Meunier (2005) and replicated the pattern reported by Morris et al. (2011), with significant priming being obtained in all three prime conditions: suffixed words (*tristesse-TRISTE*), suffixed nonwords (*tristerie-TRISTE*), and non-suffixed nonwords (*tristald-TRISTE*). We found no influence of nonword neighborhood density, with the same pattern of priming effects being seen independently of the orthographic similarity of nonword targets to real words, and no effect of affix frequency. Importantly, however, we found that non-suffixed nonword priming interacted with the language proficiency of our participants, with high proficiency participants showing greater priming than low-proficiency participants (see Fig. 1). Thus, high proficiency participants showed robust priming in all three prime conditions (the Morris et al. pattern), whereas low proficiency participants showed significantly reduced non-suffixed priming compared to the two suffixed conditions (the Longtin & Meunier pattern).

One possible explanation for the influence of language proficiency on complex nonword priming, found in the present study, is that participants with higher levels of language proficiency are more expert in mapping sublexical orthography onto whole-word orthographic representations, and would therefore rely to a lesser extent on morphological segmentation processes when processing complex stimuli. Less proficient participants, on the other hand, would be less efficient in mapping sublexical orthography onto whole-word representations, leading to less activation of the whole-word representations of embedded stems. This less efficient 'letter-word' mapping in low proficiency individuals would be compensated by a greater reliance on morpho-orthographic segmentation driven by the processing of affixes (Diependaele, Morris, Serota, Bertrand, & Grainger, 2013; Grainger & Ziegler, 2011). More efficient letter-word mapping in high proficiency individuals would lead to activation of the embedded target word during prime processing, independently of whether or not it is accompanied by a suffix. In low proficiency readers, on the other hand, embedded stem activation would not be great enough to produce priming effects without the help of an affix-stripping mechanism.

Why then did our high proficiency participants not show greater priming from complex word primes compared with complex nonword primes? Here we argue that with complex word primes, the benefits of morpho-semantics driven by whole-word processing are countered by the cost of the prime being a real word that competes with the target word for recognition. This competition does not arise with nonword primes. Had we compared complex word primes (*painter-PAINT*) with pseudo-complex word primes (*corner-CORN*), then we would have expected to observe a greater influence of semantic transparency in the more proficient readers. Such a pattern was indeed found by Andrews and Lo (2013), but only for participants who had comparatively high vocabulary scores compared with their spelling ability. On the other hand, participants with relatively good spelling ability compared with vocabulary showed comparable effects of true morphological primes and pseudo-morphological primes. Andrews and Lo (2013) interpreted these findings in terms of a 'semantic profile' (analogous to our whole-word processing bias) for the high vocabulary participants, and an 'orthographic profile' (analogous to our morpho-orthographic processing bias) for the good spellers. In the present study we failed to find an influence of spelling-vocabulary differences on priming effects. However, when analyzed separately, vocabulary interacted the most with priming, and particularly in the non-suffixed condition (VocabZ: p=.025; SpellZ: p=.087). Therefore, in a study that would better dissociate spelling and vocabulary scores (i.e., with greater variability in the difference scores), we expect the high vocabulary participants to show the pattern seen with high proficiency participants in the present study, and the good spellers to show the pattern seen with the low proficiency group.

It might be asked why previous morphological priming studies (Rastle & Davis, 2008) have failed to find evidence for embedded stem priming with word primes (*cashew-CASH*), although the embedded word identification hypothesis would clearly predict that *cash* in *cashew* is mapped onto the lexical representation of *cash*. In the light of the present results, one explanation for the absence of *cashew-CASH* priming is that previous morpho-orthographic priming experiments probably involved participants with mixed levels of language proficiency. Since non-suffixed priming is more significant in participants with high language proficiency, the *cashew-CASH* effect may have been washed out by the presence of low proficiency participants. However, we suspect that the key difference with respect to complex nonword priming is that *cash* is embedded in a letter string which itself is a word. The simultaneous activation of the lexical representations of *cash* and *cashew* generates competition between the two lexical entries, and prevents stem-target priming. Of course, lexical competition would also arise during pseudo-morphological priming (e.g., between *corn* and *corner*), but here the activation of the target word benefits from the morphological structure of the prime, which explains why stem-target priming does typically arise in this condition (e.g., Rastle & Davis, 2008). In contrast, if the embedded word is part of a string which itself is *not* a word (*tristald*), the embedded word (*triste*) is able to strongly activate the corresponding lexical representation without competition, thus producing priming. Hence, the embedded word activation mechanism provides a straightforward explanation for the priming effect in the non-suffixed condition (see also Morris et al., 2011). One clear prediction of this account is that the frequency of the word prime should determine the size of embedded word priming effects, with stronger priming being obtained with low-frequency primes such as *cashew* compared with more frequent primes such as *window* (as a prime for *WIND)*.

In the present study there was no influence of suffix frequency on the size of complex nonword priming. Although it has been hypothesized that affix frequency might be a key factor influencing the acquisition of morphological knowledge in children (e.g., Beyersmann, Castles, & Coltheart, 2012; Rastle & Davis, 2008), permitting the discrimination between typically higher frequency morphemic affixes (e.g., *er* in *corner*) and lower frequent non-morphemic units (e.g., *ew* in *cashew*), very little evidence exists to date suggesting that affix frequency affects rapid, visual word recognition in adults. Perhaps, as already discussed by Rastle and Davis (2008), a more critical difference between morphemic and non-morphemic units may be that affixes occur in combination with stem morphemes. The identification of embedded stems would thus assist during efficient morphological chunking (e.g., Brent, 1997; Brent & Cartwright, 1996), which is coherent with the here presented evidence for stem-target priming.

An interesting extension of our study in future research could involve testing of whether orthographic and morpho-orthographic priming are modulated by language proficiency in the same way. If orthographic priming *is* modulated by language proficiency in the same way, it can be concluded that embedded stem priming effects are based on purely orthographic letter-word mappings, whereas otherwise it would suggest that the observed modulation is due to the morphological chunking of embedded stems. Finally, one important remaining question concerns whether or not embedded word priming only occurs with words embedded at word-beginnings. Research from monomorphemic word recognition (Bowers, Davis, & Hanley, 2005) suggests that embedded words are identified independently of position (*hatch, drama, howl*). Moreover, studies exploring positional constraints during polymorphemic word recognition have revealed that while suffixes are identified only when they appear at word endings (Crepaldi, Rastle, & Davis, 2010), the recognition of embedded stems in pseudo-compound words (e.g., *moonhoney*) appears to be position-independent (Crepaldi, Rastle, Davis, & Lupker, 2013). However, it is yet to be established if similar principles apply to the processing of prefixed words and nonwords, although one would expect this to be the case given that embedded stems can appear at different positions in affixed words. A logical follow-up to the present study would therefore be the exploration of priming from prefixed nonwords.

Acknowledgments This project was supported by a postdoctoral grant from the Fyssen Foundation to Elisabeth Beyersmann and a grant from the European Research Council (ERC grant 230313) to Jonathan Grainger. This research was carried out within the Labex BLRI (ANR-11-LABX-0036) and has benefited from support from the French government, managed by the French National Agency for Research (ANR), Investments of the Future A*MIDEX (ANR-11-IDEX-0001-02). We also thank Jean-Baptiste Melmi for help with participant recruitment and testing.

Appendix 1

Stimuli including high-frequency suffixes:

Suffixed word prime	*Suffixed nonword prime*	*Non-suffixed nonword prime*	*Unrelated prime*	*Target*
cachette	cacheur	cachenie	perchoir	cacher
fillette	filleur	fillenie	comédien	fille
noisette	noisage	noisenie	adoption	noix
pochette	pocheur	pochenie	traction	poche
buvette	buvion	buvenie	manillon	boire
cerisier	cerisage	cerisire	batteur	cerise
fermier	fermion	fermire	nombreux	ferme
poirier	poirage	poirire	lumineux	poire
pommier	pommeur	pommire	grognon	pomme
saladier	saladion	saladire	utilisable	salade
sauteur	sautette	sautide	liaison	saut
boxeur	boxier	boxide	veston	boxe
chasseur	chassion	chasside	pompier	chasse
largeur	largier	largide	puceron	large
pêcheur	pêchette	pêchide	aileron	pêcher
million	millage	millert	laverie	mille
punition	punitage	punitert	mouchoir	punir
décoration	décoratier	décoratert	préférable	décorer
attention	attentier	attentert	fusillade	attendre
correction	correctette	correctert	plongeon	corriger

Suffixed word prime	*Suffixed nonword prime*	*Non-suffixed nonword prime*	*Unrelated prime*	*Target*
couchage	couchier	couchuse	orageux	coucher
grillage	grilleur	grilluse	champion	grille
paysage	paysette	paysuse	sorcier	pays
plumage	plumette	plumuse	sentier	plume
tournage	tournion	tournuse	grattoir	tourner

Stimuli including low-frequency suffixes:

Suffixed word prime	*Suffixed nonword prime*	*Non-suffixed nonword prime*	*Unrelated prime*	*Target*
séchoir	séchesse	séchuor	émotion	sécher
arrosoir	arrosesse	arrosuor	olympien	arroser
lavoir	lavien	lavuor	pliage	laver
trottoir	trottade	trottuor	infusion	trotter
plongeoir	plongien	plonguor	barricade	plonger
tristesse	tristerie	tristald	direction	triste
jeunesse	jeunerie	jeunald	révision	jeune
sagesse	sagerie	sagald	dormeur	sage
vitesse	viterie	vitald	mangeur	vite
hôtesse	hôtade	hôtald	tension	hôte
baignade	baignien	baignabe	voyageur	baigner
glissade	glissien	glissabe	chanteur	glisser
promenade	promenoir	promenuto	historien	promener
balade	balarie	baluto	plantoir	balader
orangeade	orangeoir	orangeabe	sélection	orange
gardien	gardoir	garduto	finesse	garde
magicien	magicesse	magicuto	pliure	magie
musicien	musicoir	musicabe	sonnette	musique
pharmacien	pharmacade	pharmacabe	tartelette	pharmacie
mécanicien	mécanicesse	mécanicuto	bijouterie	mécanique
sonnerie	sonnade	sonnebli	tricheur	sonner
épicerie	épicesse	épicebli	baladeur	épice
imprimerie	imprimerade	imprimebli	végétarien	imprimer
mairie	mairoir	maibli	joyeux	maire
boulangerie	boulangien	boulangebli	californien	boulangère

Appendix 2

Spelling dictation test

Instructions Target words are read out aloud individually by the experimenter and repeated only once. The participant's task is to write down the words. All words that can be derived from the spoken word form must be counted as correct responses.

Target words (correct responses in parentheses)

1.	phobique	(phobique, phobiques)
2.	écervelé	(écervelé, écervelée, écervelés, écervelées)
1.	phobique	(phobique, phobiques)
3.	astreint	(astreint, astreints, astreins)
4.	endolori	(endolori, endolorie, endoloris, endolories, endolorit)
5.	invaincu	(invaincu, invaincue, invaincus, invaincues)
6.	carcéral	(carcéral, carcérale, carcérales)
7.	gaillard	(gaillard, gaillards)
8.	éphémère	(éphémère, éphémères)
9.	fougasse	(fougasse, fougasses)
10.	affûtage	(affûtage, affûtages)
11.	tourteau	(tourteau, tourteaux)
12.	cirrhose	(cirrhose, cirrhoses)
13.	saucière	(saucière, saucières)
14.	dépeçage	(dépeçage, dépeçages)
15.	ganglion	(ganglion, ganglions)
16.	abattant	(abattant, abattants)
17.	oisillon	(oisillon, oisillons)
18.	importun	(importun, importuns)
19.	jouissif	(jouissif, jouissifs)
20.	offrande	(offrande, offrandes)

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